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Studierichting: Informatica
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$$z^3 = -4 - 4i\sqrt{3}$$

$$r = \sqrt{(-4)^2 + (-4\sqrt{3})^2} = 8$$

$$\theta = \frac{4\pi}{3}$$

$$-4 - 4i\sqrt{3} = 8 \operatorname{cis} \frac{4\pi}{3}$$

$$z_k = \sqrt[3]{8} \operatorname{cis} \left(\frac{\frac{4\pi}{3} + k2\pi}{3} \right)$$

$$= 2 \operatorname{cis} \left(\frac{4\pi}{9} + k\frac{2\pi}{3} \right)$$

$$z_0 = 2 \operatorname{cis} \frac{4\pi}{9}$$

$$z_1 = 2 \operatorname{cis} \frac{10\pi}{9}$$

$$z_2 = 2 \operatorname{cis} \frac{16\pi}{9}$$

References